

1. Refactoring techniques used

In this task, I used several small refactoring techniques to improve the code structure without changing its behaviour. The main ones were Extract Function, Rename Variable, Inline Variable, and Parametrise Function. These changes were intended to make the code easier to read and easier to modify later.

2. What I changed and why

The main part I refactored was `generate_report()`. In the original version, it handled too many things in one function, including looping through customers, calculating totals, applying tax rules, and printing customer status messages. I split this into smaller helper functions such as `_calculate_total_spent()`, `_apply_tax_if_needed()`, `_print_discount_status()`, and `_print_priority_status()`. This made the logic clearer and easier to follow.

I also renamed unclear variables like `x`, `y`, `z`, and `a` to more meaningful names such as `customer_name`, `purchases`, `purchase`, and `total_spent`. In the shipping fee part, I removed unnecessary temporary variables and reduced repeated logic by introducing one helper function that takes parameters. This made the code less repetitive and more consistent.

3. How I checked behaviour preservation

Before refactoring, I ran the provided unit tests. After making the changes, I ran the same tests again. All 7 tests passed before and after the refactoring. I also reran coverage to make sure the test execution still worked as expected. This gave me confidence that the observable behaviour had not changed.

4. Benefits and drawbacks

The main benefit of the refactoring is better readability. The code is now easier to understand because the responsibilities are split into smaller and clearer parts. I think this would also make future changes easier.

A possible drawback is that refactoring still carries some risk. Even small changes can break existing behaviour if the tests are not strong enough. It can also take time and may cause conflicts when several people are working on the same files.

5. Reflection

This exercise helped me understand that refactoring is not just about shortening code. It is mainly about improving structure and making responsibilities clearer while keeping the program behaviour the same. Even small changes can make the code much easier to read and maintain.